

EDU GEVIE - GOOGLE GEMINI POWERED LEARNING ASSISTANT

Edu Gevie – Google Gemini Powered learning Assistant

Documentation

Definition:

EDU GEVIE – Google Gemini Powered Learning Assistant is an **AI-based educational platform** that uses **Google Gemini technology** to help students learn more effectively. It provides personalized learning support such as **answering questions, explaining difficult concepts, summarizing study materials, generating quizzes, and creating study plans.**

Team members Name & Roles:

1.Name: SangaMithra M (Team Leader)

Role: Voice over

2.Name: Pooja Sri V (Team Member)

Role: Documentation

3.Name: Sharmi K (Team Member)

Role: Demo video

4.Name: Yashmin G (Team Member)

Role: Coding

5.Name: Shobana P (Team Member)

Role: Testing & Review.

1.0 AI Doubt Solver is an artificial intelligence feature that allows students to ask questions or clear their doubts and receive instant, understandable explanations. In EDU GEVIE, Google Gemini can analyze the student's question and provide a suitable answer.**Solver – Definition**

An AI Doubt

1.1 Example

Student: “What is photosynthesis?”

AI Doubt Solver: “Photosynthesis is the process by which green plants make their food using sunlight, carbon dioxide, and water. Oxygen is released as a by-product.”

1.2 Objective of AI Doubt Solver

The main objective of the **AI Doubt Solver** in EDU GEVIE is to **provide students with quick, accurate, and easy-to-understand answers to their academic questions using AI.**

Key objectives:

1. To clear students' doubts instantly.
2. To provide step-by-step explanations.
3. To make difficult concepts easier to understand.
4. To support personalized learning.
5. To reduce students' dependency on external help.
6. To provide 24/7 learning assistance.

2.0 Personalized Learning:

1. **Personalized learning is an AI-based approach that** adapts the learning content, difficulty level, pace, and study materials according to each student's individual needs and performance.
2. **To help students learn more effectively by providing** customized educational content and guidance **based on their strengths, weaknesses, and learning progress.**

2.1 Objective:

1. Learner-centered approach – **Focuses on the individual learner's needs, interests, and abilities.**
2. Individual learning pace – **Students can learn at a pace suitable for them.**
3. Personalized goals – **Learning objectives are adapted to each student.**
4. Adaptive learning – **Content and activities change according to the learner's performance.**
5. Learner choice – **Students may have choices in topics, resources, or learning activities.**
6. Continuous assessment – **Regular assessment helps identify strengths and areas for improvement.**
7. Immediate feedback – **Learners receive timely feedback to improve their performance.**
8. Use of technology – **Digital tools and learning platforms can support personalized instruction.**
9. Teacher support – **Teachers guide, monitor, and modify learning experiences according to student needs**

2.2 Example:

If a student performs poorly in **Algebra**, EDU GEVIE can identify the weakness and provide:

- Easy explanations of Algebra concepts
- Additional practice questions
- Personalized quizzes
- Step-by-step solutions
- A study plan focused on Algebra

3.0 Notes & Summarization

Notes & Summarization is an AI feature that **converts lengthy study materials, documents, or lessons into short, clear, and easy-to-understand notes** while keeping the important information. To help students **save time, understand key concepts quickly, and revise lessons more easily.**

Note-making is the process of writing down the **main and important points** from a passage, lecture, or topic in a short and organized form. **Summarization** is the process of presenting the **main ideas of a longer text in a shorter form**, using your own words.

Example:

A student uploads a **20-page Biology chapter**. EDU GEVIE analyzes the chapter and generates:

- Important points
- Key definitions
- Main concepts
- Short notes

3.1. Objectives

- To understand the main ideas.
- To save time while studying.
- To remember important information.
- To organize information systematically.
- To make revision easier.
- To remove unnecessary details.
- To improve reading and writing skills.

3.2 Important Features

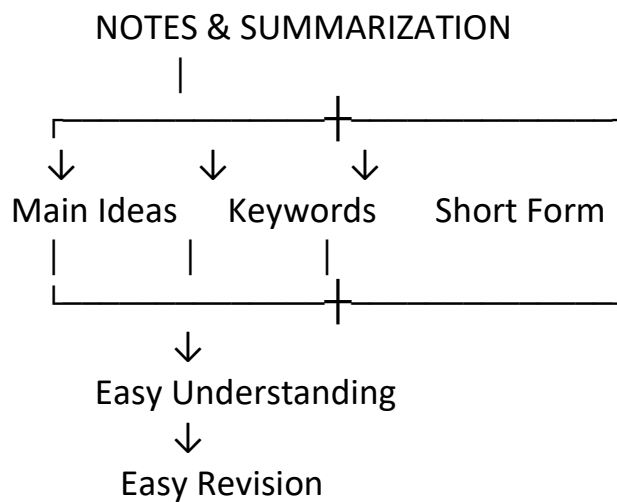
- **Short and concise**
- **Clear and organized**
- Uses **headings and subheadings**
- Includes **keywords and important facts**
- Uses **abbreviations and symbols**
- Avoids unnecessary examples and repetition.

3.3 Example

Original Passage:

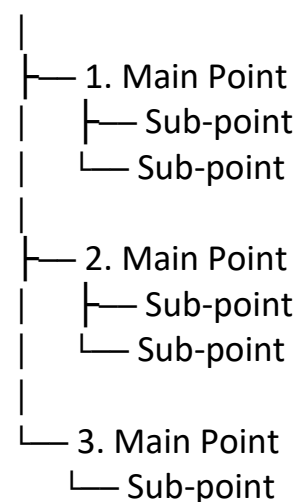
“Trees are very important for life. They provide oxygen, absorb carbon dioxide, prevent soil erosion, provide shade, and offer shelter to many animals. Therefore, protecting trees is essential for maintaining environmental balance.”

3.4 Simple Diagram

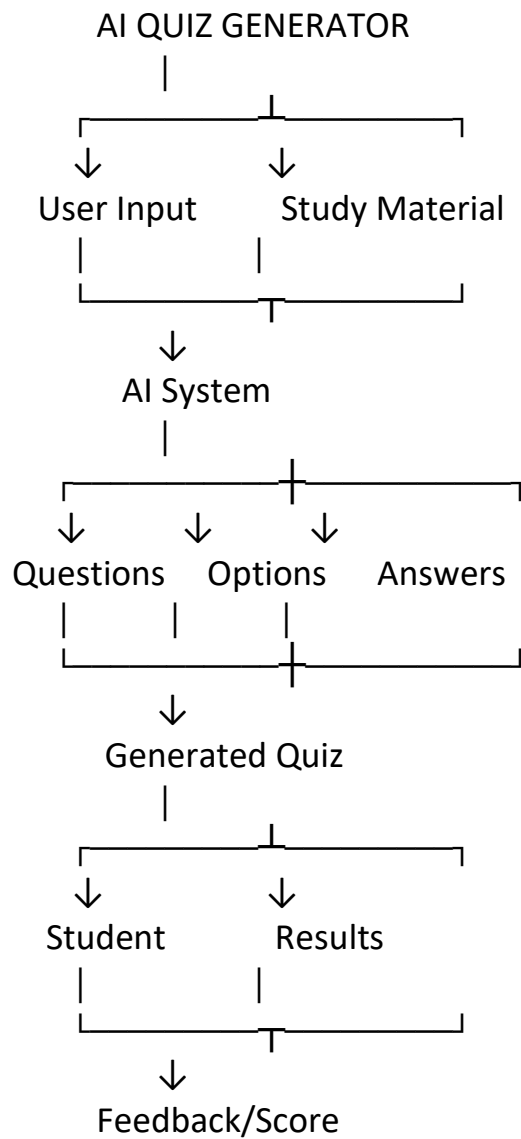


3.5 Basic Format of Notes

TITLE



4.0 AI Quiz Generator :



4.1 Output of AI Quiz Generator

Input: Topic, notes, textbook content, or study material.

Output:

- Multiple-choice questions (MCQs)
- True/False questions
- Multiple-answer questions
- Correct answers
- Hints
- Explanations/feedback
- Final score and results

Example Output:

Topic: Notes & Summarization

Q1. What is the main purpose of note-making?

- A. To record every word
- B. To organize important information
- C. To make a passage longer
- D. To repeat information

Answer: B

Q2. A summary presents the main ideas in a shorter form.

True / False

Answer: True

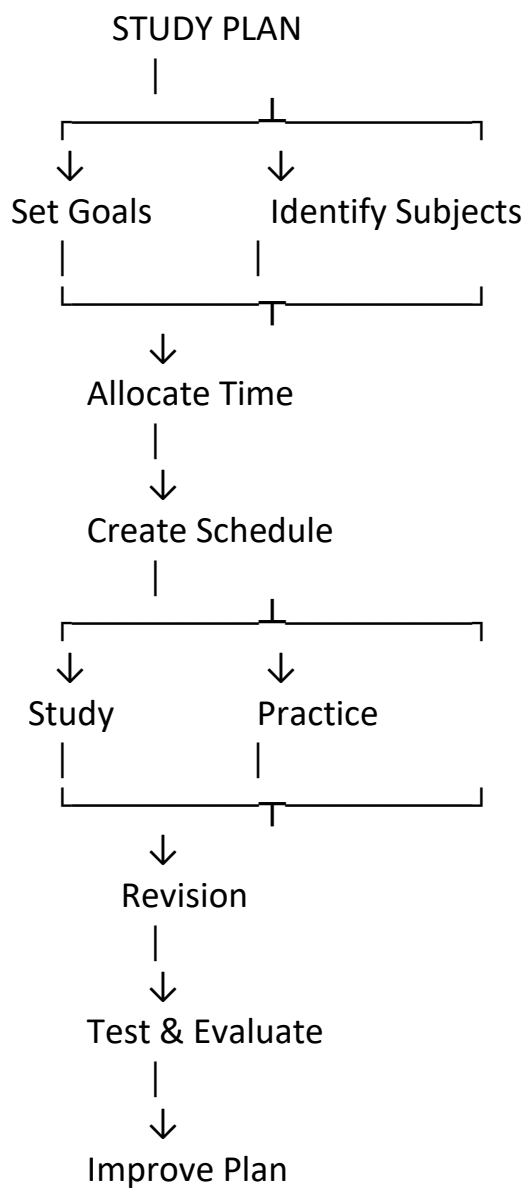
5.0 Study Plan:

A **study plan** is a structured schedule that organizes **what to study, when to study, and how much time to spend** on each subject or topic.

Objectives

- Organize study time effectively.
- Cover all subjects systematically.
- Set clear learning goals.
- Improve concentration and consistency.
- Provide time for revision and practice.
- Reduce last-minute studying.

5.1 Study Plan Diagram



5.2 Example

Time	Activity
6:00–7:00 AM	Mathematics
7:00–7:30 AM	Revision
5:00–6:00 PM	Science
6:00–6:30 PM	Break
6:30–7:30 PM	English
8:00–8:30 PM	Practice Questions

Key Points

- Set **realistic goals**.
- Follow a **regular timetable**.
- Include **short breaks**.
- Revise previously studied topics.
- Practice with questions and tests.
- Review your progress and adjust the plan.

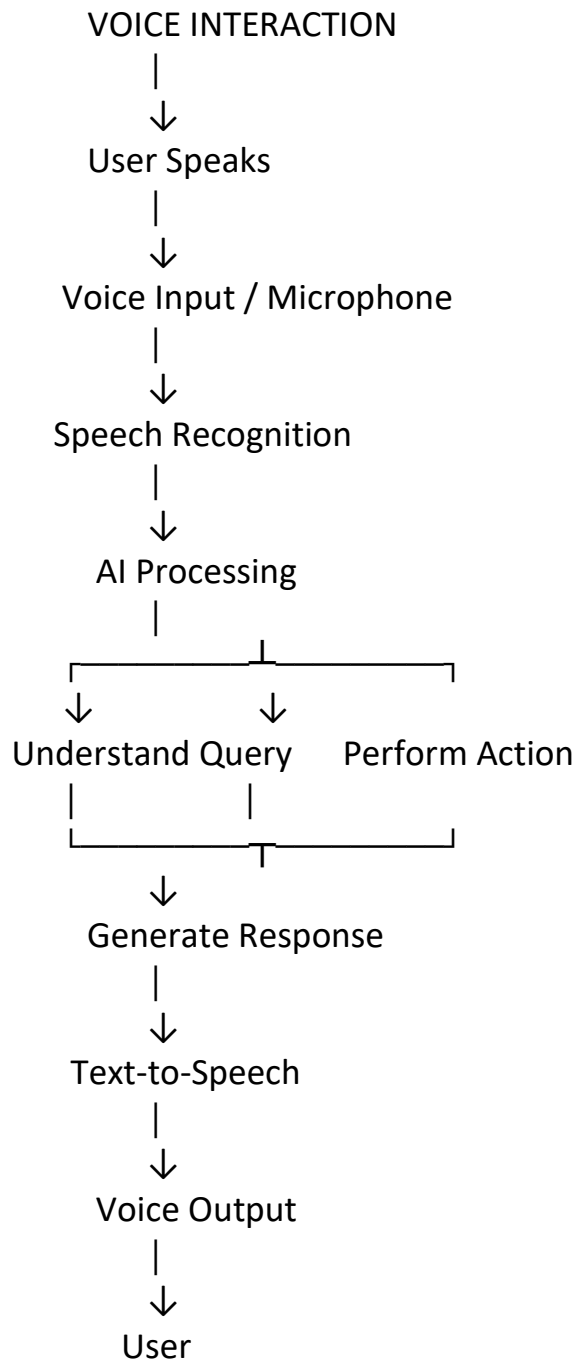
6.0 Voice Interaction:

Voice interaction is a method of communication in which a user interacts with a computer or AI system through **spoken commands and responses** instead of typing.

Objectives

- Enable hands-free interaction.
- Make technology easier to use.
- Support natural communication.
- Provide quick responses.
- Improve accessibility.
- Allow users to give commands using speech.

6.1 Voice Interaction Diagram



Examples

- Voice assistants
- AI chatbots with speech
- Voice-controlled smart devices
- Speech-to-text applications
- Voice-based search

6.2 Input and Output

Input: User's spoken command or question.

Processing: Speech recognition → AI understanding → response generation.

Output: Spoken answer, information, or completed action.

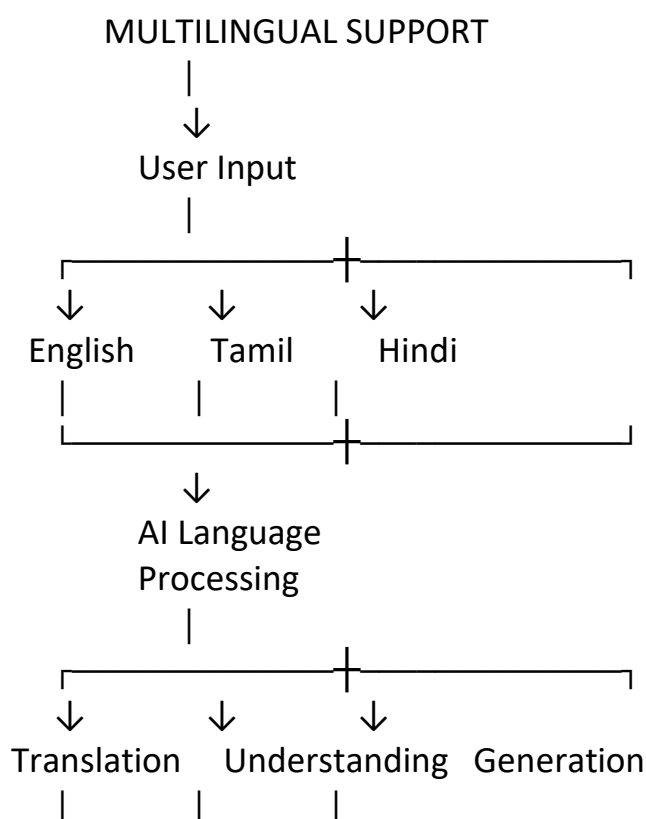
7.0 Multilingual Support:

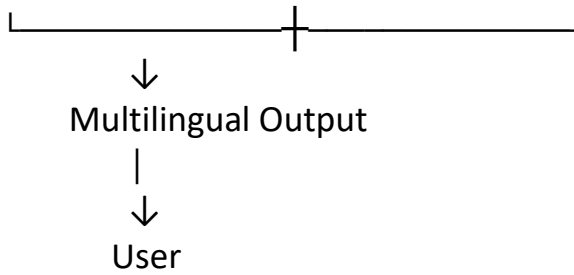
Multilingual support is the ability of an AI system or application to **understand, process, and communicate in multiple languages**.

Objectives

- Support users from different language backgrounds.
- Make information accessible in multiple languages.
- Enable translation between languages.
- Improve communication and understanding.
- Help users learn and use different languages.
- Provide a more inclusive user experience.

7.1 Multilingual Support Diagram





Examples

- Translating English into Tamil or Hindi.
- Chatbots that answer questions in multiple languages.
- Voice assistants supporting different languages.
- Multilingual educational applications.
- Websites and apps with multiple language options.

8.0 Progress Tracking:

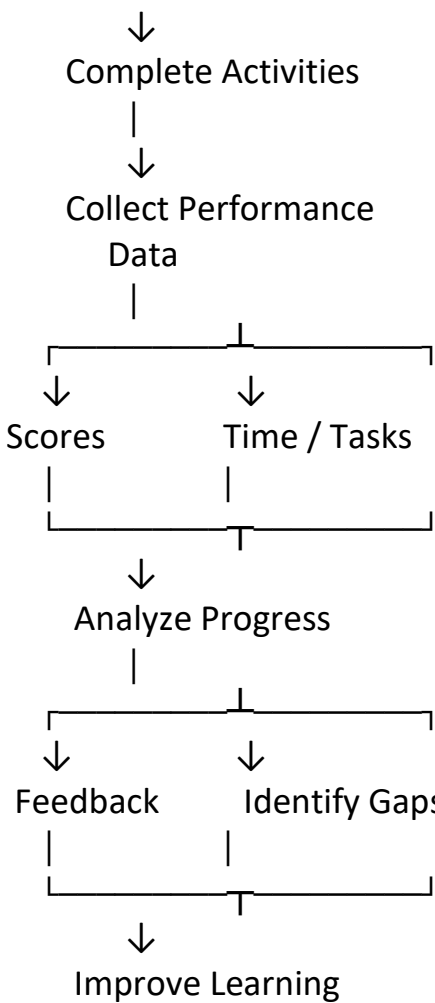
Progress tracking is the process of **monitoring and recording a learner's performance, activities, and improvement over time.**

Objectives

- Monitor learning progress.
- Identify strengths and weaknesses.
- Track completed topics and tasks.
- Measure performance through scores and results.
- Motivate learners to achieve their goals.
- Identify areas that need improvement.
- Help create better study plans.

8.1 Progress Tracking Diagram





Examples

- Tracking quiz scores.
- Recording completed lessons.
- Monitoring study hours.
- Showing percentage of syllabus completed.
- Tracking strengths and weak topics.
- Providing progress reports.

8.2 Input and Output

Input: Quiz scores, completed lessons, study time, assignments, and activity data.

Processing: Collect → analyze → compare → evaluate.

Output: Progress percentage, scores, reports, feedback, and improvement suggestions.

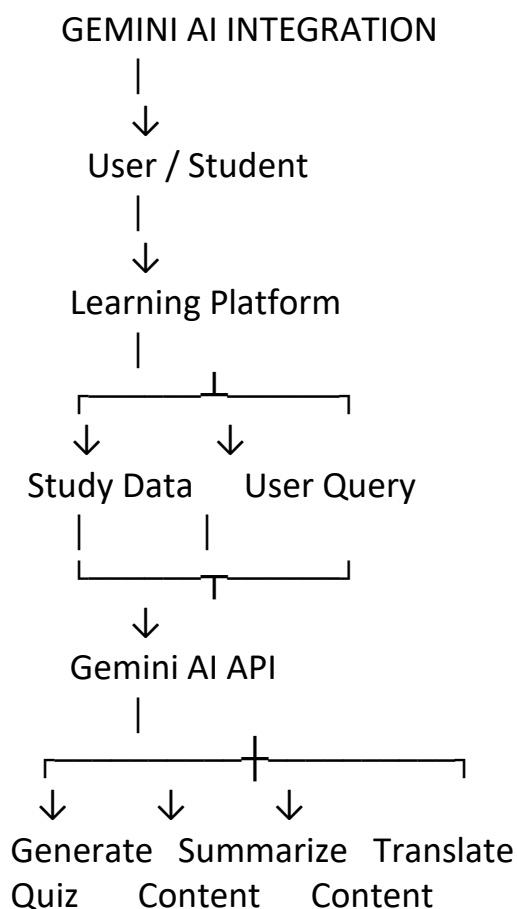
9.0 Gemini AI Integration:

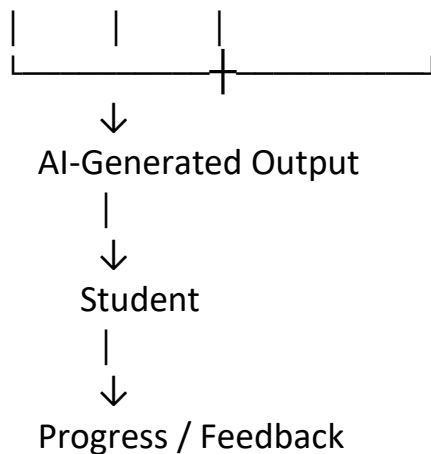
Gemini AI integration is the process of connecting an application or educational system with **Google Gemini's AI capabilities** to provide intelligent features such as question generation, summarization, translation, and personalized assistance.

Objectives

- Generate educational content automatically.
- Create quizzes and practice questions.
- Summarize study materials.
- Provide AI-based explanations.
- Support multiple languages.
- Give personalized learning assistance.
- Analyze learner responses and provide feedback.

9.1 Gemini AI Integration Diagram





9.2 Applications

- **AI Quiz Generator:** Creates questions from study material.
- **Notes & Summarization:** Converts long content into concise notes.
- **Study Plan:** Helps organize learning activities.
- **Voice Interaction:** Supports conversational learning.
- **Multilingual Support:** Provides content in different languages.
- **Progress Tracking:** Analyzes learning performance and provides feedback.

9.3 Input and Output

Input: Questions, notes, textbook content, voice commands, or learning data.

Processing: Application → Gemini AI → AI analysis/generation.

Output: Answers, quizzes, summaries, translations, explanations, and feedback.

10.0 Student-Friendly Interface:

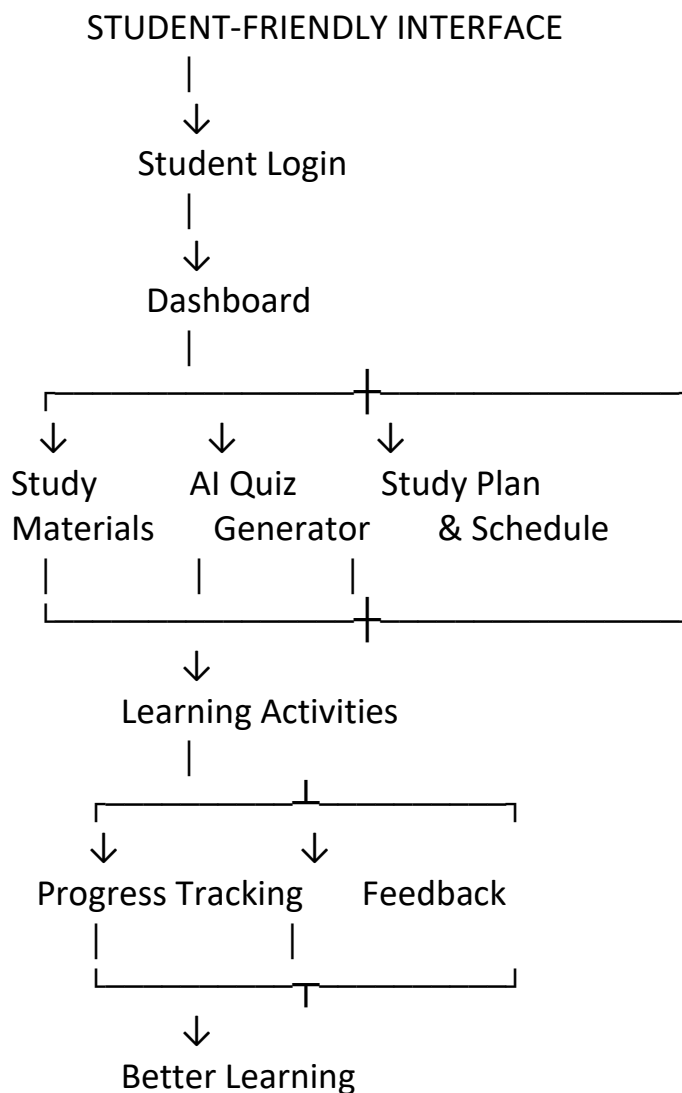
A **student-friendly interface** is a simple, clear, and easy-to-use design that helps students **access learning materials, use educational tools, and track their studies comfortably.**

Objectives

- Make the application easy to understand and use.
- Provide simple and clear navigation.
- Reduce confusion while using learning features.
- Give quick access to study materials.

- Display progress clearly.
- Support students with different learning needs.
- Provide an engaging and comfortable learning experience.

10.1 Student-Friendly Interface Diagram



Key Features

- Simple navigation
- Clear buttons and menus
- Responsive design
- Easy-to-read content
- Interactive quizzes
- Progress dashboard
- Voice interaction
- Multilingual support

10.2 Input and Output

Input: Student actions, questions, study preferences, and learning data.

Processing: Interface receives the input and connects the student with relevant learning features.

Output: Study materials, quizzes, AI responses, progress reports, and feedback.

The screenshot displays the Google Gemini web interface. The browser address bar shows 'google.com/search/studk/ee--chat'. The Gemini logo is in the top left. The main chat area shows a conversation with Gemini. The user's input is: 'Hi Gemini, can you help me create a study plan and some quiz questions for 10th Standard Science - Physics, specifically the 'Light: Reflection and Refraction' chapter? Please include important concepts, key terms, and 10 multiple-choice questions with answers.'

Gemini's response is: 'Of course! I can help you create a study plan and quiz in English for 10th Standard Physics - Light: Reflection and Refraction. Here is a structure: 10-Day Study Plan: • Day 1-2: Introduction and Properties of Light • Day 3-4: Reflection - Types of Mirrors, Focal Length • Day 5-6: Refraction - Laws, Refractive Index • Day 7-8: Lenses - Convex, Concave - Ray Diagrams • Day 9: Applications - Human Eye, Microscopes • Day 9: Applications - Human Eye, Microscopes • Day 10: Revision and Mock Test'.

Below the study plan, there are 'Practice Quiz Questions: • Day 1: • What is reflection? Explain specular vs diffuse reflection. • State the laws of refraction and the definition of refractive index. • Day 2: State laws of refraction and the definition of refractive index.'

On the right side, there is a 'Study Resource: Visuals' section with three diagrams: 'Light Reflection and Refraction', 'Light Reflection and Refraction', and 'Light Reflection and Refraction'. Each diagram shows light rays interacting with a mirror or lens.

At the bottom, there is a 'Practice Questions' input field and a 'Google Gemini basics to sratchy smiltoity anandulhuh, Google.com: Learn more' link.